

Bridge Day 12 INSTRUCTOR KEY

For Loops, Range, Counters, and Accumulators

Learning sequence: Predict - Trace - Explain - Test - Interpret

Tuesday, July 21, 2026 • 50 points • longitudinal template 07242026_v1

Name		UIN		Section	
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Daily target and independence boundary

Build fixed-count programs that preserve loop state and exact bounds.

Allowed tools	Day 10 tools plus for, range, counters, and accumulators
Support supplied	Required inputs, required outputs, and one sample run are supplied.
Student decides	Initialization, range bounds, accumulator updates, and the placement of the threshold decision.
Scaffold level	State inventory prompted once; loop body is student-authored.

Evidence and scoring

Evidence	Points	Secure work shows
Integrated trace	10	intermediate state, condition results, exact output, and meaning
Diagnosis and repair	10	actual, intended, cause, repair, and a discriminating test
Complete program and tests	30	coherent executable logic, required output, assumptions, and defended tests

Task 1. Integrated trace - 10 points

Trace the range, the conditional counter, and the accumulator after every pass.

```
total = 0
multiple_count = 0

for number in range(2, 11, 2):
    if number % 4 == 0:
        multiple_count = multiple_count + 1
    total = total + number - multiple_count
    print(number, total, multiple_count)
```

Your task

1. List the values produced by range(2, 11, 2).
2. Create one pass ledger containing number, multiple_count after the if, and total after the update.
3. Write all printed lines in exact order.

Answer 1. The range values are 2, 4, 6, 8, 10. The stop value 11 is excluded.

Answer 2. Pass states (number, multiple_count, total): (2, 0, 2), (4, 1, 5), (6, 1, 10), (8, 2, 16), (10, 2, 24). The counter update occurs before that pass contributes to total.

Answer 3. Exact output lines: 2 2 0; 4 5 1; 6 10 1; 8 16 2; 10 24 2.

Task 2. Diagnose and repair - 10 points

The intended program must add the integers from 1 through limit, inclusive.

```
limit = int(input())
total = 0

for number in range(1, limit):
    total = total + number

print(total)
```

Your task

1. For limit = 4, state the actual and intended totals and identify the exact cause.
2. Write the minimal repair and a boundary test that would expose the original bug.

Answer 1. range(1, 4) produces 1, 2, 3, so the actual total is 6. The intended inclusive total is $1 + 2 + 3 + 4 = 10$. The loop excludes limit because range excludes its stop value.

Answer 2. Minimal repair: for number in range(1, limit + 1). A strong boundary test is limit = 1: the buggy loop totals 0, but the repaired loop totals 1.

Task 3. Construct a complete program - 30 points

Program specification

- Read sample_count as an integer and threshold as a float.
- If sample_count is not positive, print invalid and do not ask for readings.
- Otherwise read exactly sample_count floating-point readings.
- Compute the total, average, and number of readings at or above threshold.
- Print Total to one decimal, Average to two decimals, and At or above as an integer on separate labeled lines.

Required planning evidence

1. Name the state that must exist before the loop and give each initial value.

Program workspace**Model program**

```
sample_count = int(input())
threshold = float(input())

if sample_count <= 0:
    print("invalid")
else:
    total = 0.0
    at_or_above = 0

    for sample_number in range(sample_count):
        reading = float(input())
        total = total + reading
        if reading >= threshold:
            at_or_above = at_or_above + 1

    average = total / sample_count
    print(f"Total: {total:.1f}")
    print(f"Average: {average:.2f}")
    print(f"At or above: {at_or_above}")
```

Reasoning: total and at_or_above must be initialized once before repetition. The loop runs exactly sample_count times because range(sample_count) produces that many indexes.

Reasoning: The division occurs after the loop and only in the positive-count branch, so division by zero is impossible. The labeled f-strings make each reported quantity unambiguous and enforce the required precision.

Task 3 continued. Test and defend - included in 30 points

Continue code if needed.

Expected tests and results

1. Count 4, threshold 4.0, readings 3.5, 4.0, 2.5, 5.0 prints Total: 15.0; Average: 3.75; At or above: 2 on separate lines.
2. Count 0 and any threshold prints invalid and requests no readings.

Required tests, expected results, and brief defense

Scoring guidance: 4 points state plan; 4 input/guard; 6 correct loop bound and repeated input; 5 total; 4 threshold count; 4 average/labeled precision; 3 tests.

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VIP Evidence Handoff

Learning sequence: Predict - Trace - Explain - Test - Interpret

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Why engineers use this page

Use deliberate range bounds and comparison state to collect multiples and preserve the smallest value.

Decision boundary: Code is useful only when its stored state, visible result, assumptions, units, limits, and tests support a defensible engineering interpretation.

Construct readiness before independent work

New authoring tools: for loop, range call, list literal as collector, append call

Carried authoring tools: assignment, print call, arithmetic expression, modulo, input call, int conversion, float conversion, f string, comparison expression, boolean operator, if elif else, abs call, counter update, while loop, loop state update, termination condition

Supplied-code exposure only: list index read

An exposure-only construct may be traced in complete supplied code, but the independent task will not require students to invent it before its authoring day.

Supplied append model before Activity 18.18

Run and trace this complete model before the transfer task. Notice that append stores one qualifying value per pass; it does not print or replace the collector.

```
limit = 12
divisor = 3
multiples = []

for number in range(1, limit + 1):
    if number % divisor == 0:
        multiples.append(number)

print(multiples)
```

Expected output: [3, 6, 9, 12]

Notebook cells and task constructs

Activity	Work	Stable cells	Required authoring constructs
18.18	Multiples In Range	18-18-work / 18-18-transfer / 18-18-cold	append call, arithmetic expression, for loop, range call
18.20	Smallest Value	18-20-work / 18-20-transfer / 18-20-cold	comparison expression, for loop, if elif else

Readiness evidence

1. Write the first value, final included value, and excluded stop value for the range used in Activity 18.18.

Model response: The exact values depend on the sample or transfer inputs, but the final desired endpoint must be included and Python's range stop must be one step beyond it when the step is positive.

2. Explain why Activity 18.20 initializes from real data instead of from zero.

Model response: Zero is not guaranteed to be in the data and can be smaller or larger than every real value. Initializing from the first data value makes the comparison valid for all nonempty inputs.

Timing and help trigger

Record a first prediction before running. If you still lack an executable plan after about 8 focused minutes, use the linked ThinkCSPy refresher or the supplied model. If two attempts fail for the same named requirement, write the exact mismatch and ask a peer mentor or instructor a targeted question. Timing is diagnostic evidence, not a speed grade. **Start or elapsed minutes:** _____ **My help trigger:** _____

Engineering Evidence Record

Complete one record from a model, transfer, or Independent Program Studio build. Generic statements are not sufficient; use actual values, a real revision, and one concrete next test.

Evidence field	Record
Prediction	A specific expected state or output before execution.
Observed Result	The actual state or output, including a value or exact text.
Revision Reason	The defect or mismatch and the change that addresses it.
Engineering Meaning	How the evidence changes or supports the engineering decision.
Units Or Not Applicable	The physical unit, or the evidence type when no unit applies.
Assumption	One condition accepted as true for this model or rule.
Model Limit	One situation the result does not justify or predict.
Next Test	A concrete boundary, invalid, or alternate case with an expected result.
Help Trigger	The exact unresolved point that would trigger a targeted question.
Minutes Used	Approximate minutes from first prediction through revision.

Notebook and submission procedure

1. Attempt, predict, run, inspect, and revise before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only those visibly labeled Studio cells are scored for independent mastery.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.